Prabhleen Gill

Reinforcement Learning

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Model-Based vs. Model-Free in Cliff Walking Environment

Abstract

This study performs an evaluation of three reinforcement learning algorithms in the context of the Cliff Walking problem. Namely, it compares a model-based algorithm, Value Iteration, to two model-free algorithms, Sarsa and Q-Learning, and measures their abilities in generating a solution to this problem. In particular, this study looks at the convergence speeds as well as the optimality of the policies to evaluate the efficiency of each algorithm. The results highlight the trade-offs between reliance on a model and exploring possibilities beyond a model. Using these results, this study explores the unique traits of each algorithm and their ideal use case to evaluate their overall performance.

Introduction

When faced with a variety of decisions, one may prefer to make a choice that results in the most gain. However, it is not always easy to make that choice, especially when there are numerous factors to consider. The Cliff Walking problem, created by Sutton and Barto in *Reinforcement Learning: An Introduction*, is an ideal representation of the weight that rewards have on decision-making. A player must make it to the goal earning the highest possible reward, with the risk of falling off the cliff and suffering extreme loss. There are numerous ways for a decision to be made, but the best decision can vary among multiple methods. In this experiment, we explore and compare some of these methods, specifically model-based and model-free, to measure their ability in making the best decision.

Environment Description

The environment used for this project was Cliff Walking from OpenAI Gym. The game takes place on a 4 x 12 grid, and the cells are numbered 0 to 47 by rows. The player starts on cell 36 and their goal is to reach the end state, cell 47, without falling off the cliffs. The cliffs are located on cells 37 to 46, and the player is able to fall off from any of the surrounding cells except the end state. When the player falls off, they return to the starting cell. There are four actions that can be taken: up, right, down, left. The player earns a reward of -1 for every safe move and -100 if they fall off the cliff. The episode ends once the player reaches the end state. The latest version of Cliff Walking allows the player to slip perpendicular to the intended direction with a 33% probability. For this experiment, this feature was disabled.

Details on Algorithms and Experiment

The algorithms used for this experiment were Value Iteration, Sarsa, and Q-Learning. Value Iteration is a model-based algorithm used for finding the optimal policy for each state. It calculates the value of each state for every action using the Bellman equation, and follows predetermined transitions and rewards. The action which gives the maximum state value is chosen as the policy for the state.

Sarsa is a model-free algorithm used for finding the values of state-action pairs to determine the optimal policy for a state. It is on-policy, meaning that it follows a policy to determine next actions. Sarsa uses the next actions of a state to calculate its value, and the best action is chosen based on an ε-greedy policy. Since it considers all possible state-action pairs, Sarsa tends to choose policies that are safer, meaning that there is a lower risk of earning lower rewards.

Like Sarsa, Q-learning is a model-free algorithm used for finding the values of state-action pairs to determine the optimal policy for a state. However, it is off-policy, meaning that it does not follow a policy to choose next actions. Instead, when it calculates the value of a state-action pair, Q-Learning considers the maximum state-action value of the next state and all of its possible actions. Sarsa considers just the state-action value of the next state and action. Since Q-Learning considers the maximum values, it chooses policies that are more optimal in terms of total rewards, meaning that there is still a risk of earning lower rewards by following the policy.

For the experiment, the Cliff Walking environment from OpenAI Gym was used without additional customizations. Each algorithm was run separately and results were displayed on a heatmap to highlight the difference in values. The optimal policy for each state was also displayed on the heatmap through arrows to better visualize the ideal path determined by the algorithm. The figures were generated using Matplotlib and Seaborn. Numpy was used within the algorithms to initialize the value tables and apply the built-in greedy policy. Sarsa and Q-Learning were run for 300 episodes, since they had converged within that range.

For the model-free algorithms, epsilon, alpha, and gamma were the hyperparameters used in calculating the values for each state-action pair. The respective values for each hyperparameter were 0.1, 0.1, and 0.9. A lower epsilon was chosen to focus primarily on choosing the best option while also ensuring some exploration. It was important to have exploration to consider options that produced lower risks of earning lower rewards. A lower alpha was chosen to increase variance by slowing down learning. Learning too quickly can result in failure to converge and fluctuations in Q-values, which may produce inaccurate results. For similar reasons, a higher gamma was chosen to motivate long-term planning.

For value iteration, gamma and theta were the hyperparameters used, and their respective values were 0.9 and 0.0001. Similar to the model-free algorithms, a higher gamma was chosen to focus on future rewards. A lower theta was chosen to also slow down the learning process by increasing the number of iterations. Using fewer iterations may result in a policy that is not the most optimal.

Results and Analysis

Figures 1 and 2 display the learned Q-values of Sarsa and Q-learning after running 300 episodes. Both algorithms produced a policy that took the player from start to finish, but Sarsa produced a safer path while Q-Learning produced the optimal shortest path. Q-Learning travels along the edge of the cliff as it is the shortest path to get from start to finish. Sarsa goes one row up and travels through the middle of the grid until the last column, then travels down to the finish. Sarsa had a slightly larger range of Q-values than Q-Learning, and this could be attributed to the exploration brought on by the ε-greedy policy. Q-Learning used a greedy policy to calculate Q-values, resulting in little to no exploration.

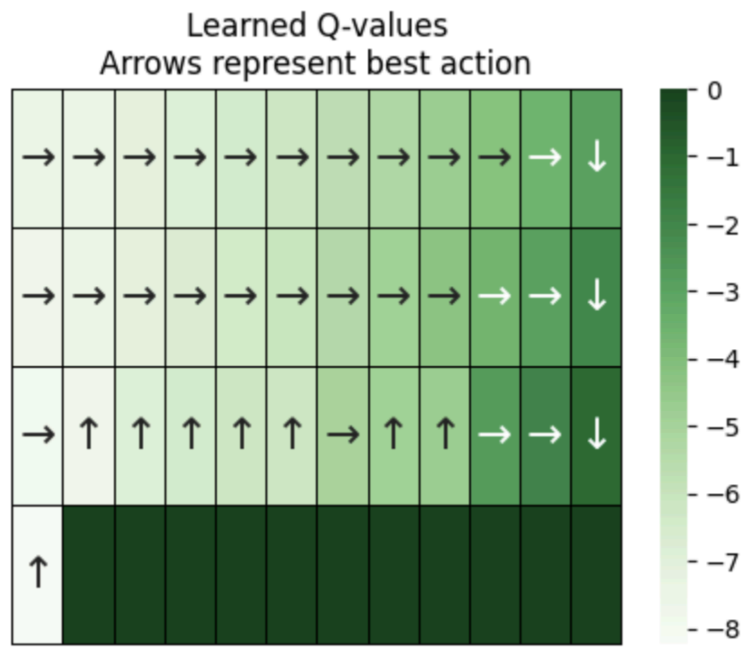

Figure 1. Learned Q-Values of Sarsa

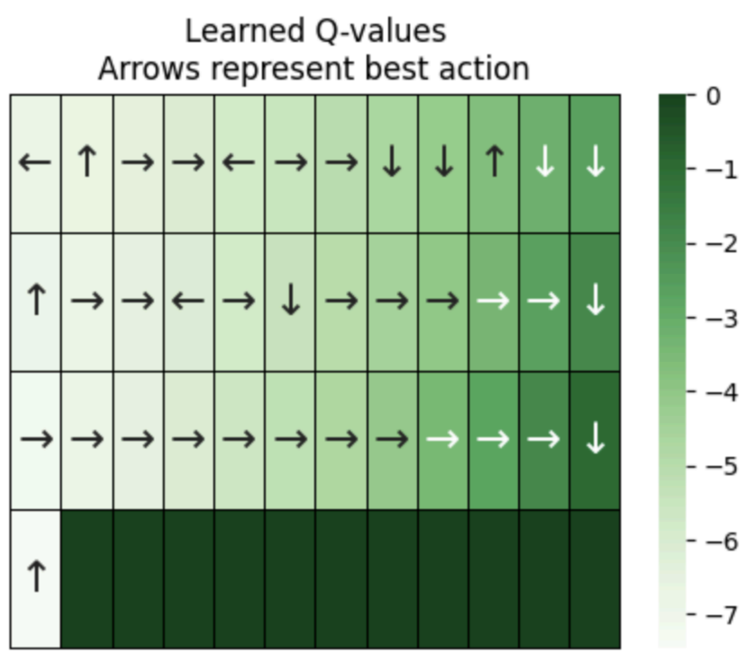

Figure 2. Learned Q-Values of Q-Learning

Figures 3 and 4 depict the sum of rewards per episode for Sarsa and Q-Learning. Both algorithms demonstrated an extreme drop in rewards at episode 2, but grew steadily to episode 50. Q-Learning appeared to have more fluctuations in rewards as the episodes progressed, while the rewards of Sarsa remained more stable with occasional drops. This could be due to the greedy policy used by Q-Learning, which prioritized optimality over safety and thus resulted in numerous falls off the cliff. Since Sarsa used an ε-greedy policy, it was able to explore the actions of states farther from the cliffs. Overall, it is apparent that Sarsa was able to yield more rewards on average compared to Q-Learning.

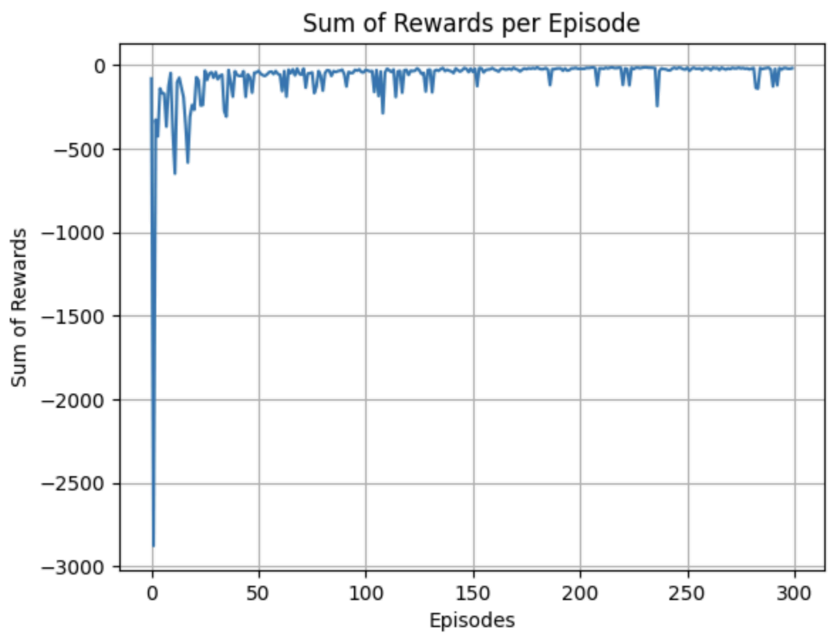

Figure 3. Rewards per Episode of Sarsa

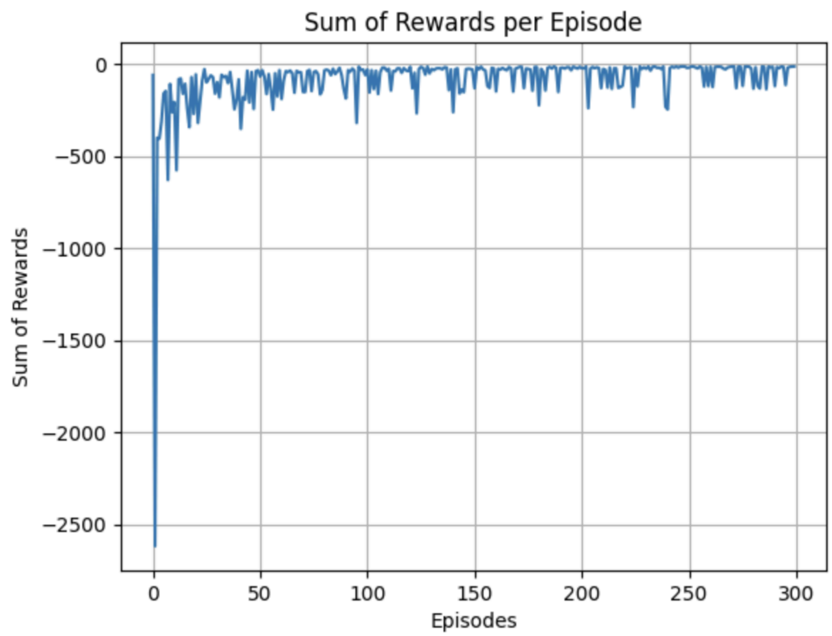

Figure 4. Rewards per Episode of Q-Learning

Figure 5 depicts the learned V-values of Value Iteration. The algorithm was able to converge after 15 iterations. The arrows point in the direction closest to the end state, which indicates that learning was able to occur at an ideal rate. The V-values appear to be similar to the Q-values produced by the model-free algorithms. Because Value Iteration is a model-based algorithm, it relies on the given model instead of creating its own model using samples from episodes. This saves time in determining the state values as the algorithm can simply use what is given. While the end results are similar to those of model-free algorithms, using a model-based algorithm may prove to be more beneficial in environments with unchanging transitions and rewards.

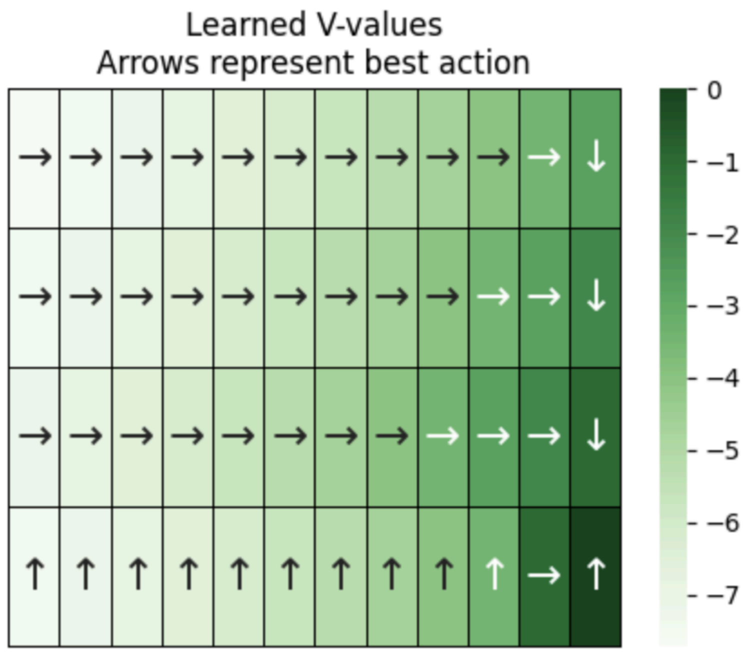

Figure 5. Learned V-Values of Value Iteration

The model-based algorithm gave a general direction of how to proceed through the grid, while the model-free algorithms gave specific paths to take to reach the end. This is because the model-free algorithms explored state-action pairs rather than just states, meaning that a wider range of possible cases were considered. The differences in results highlights the unique purposes of each type of algorithm. Although both produce the optimal policy for each state, model-based algorithms could be more useful in a case where a planned route is needed to complete a task. Model-free algorithms could be more useful in cases where end rewards hold more significance and exploration is encouraged. Such cases occur in environments that do not remain fixed and focus more on efficiency of a task rather than completion. In particular, Sarsa would be more suitable for cases where learning is prioritized, while Q-Learning would be better for cases where performance is prioritized.

Conclusion and discussion

The Cliff Walking environment modeled the problem of efficient decision making. Based on the results of this experiment, model-based and model-free algorithms could be applied to this problem to generate solutions that are logical and accurate. Concurrently, the results also highlighted the distinct use cases for each type of algorithm: whether an environment remains fixed, and whether learning and end rewards hold more significance. This study found that these factors are important to consider when choosing a reinforcement learning algorithm, as the differences in results were attributed to the unique nature of each algorithm.

Although the unique nature of each algorithm plays a key role in its performance, other important factors to consider are the nature of the environment and hyperparameters. For this experiment, is_slippery was disabled for simplicity; however, the results may have differed due to the probability of going into the unintended direction. Value Iteration may not have performed as well as it did with is_slippery disabled since model-based algorithms must rely on a model with fixed transitions and rewards. For this environment, probabilities would have to be manually adjusted for the model-based algorithm to perform. For the model-free algorithms, more episodes would be required for convergence as the changing nature of the environment encourages more exploration. Since learning would become the greater focus, Sarsa would perform better than Q-Learning.

Hyperparameters also played an important role in producing the results. A lower epsilon was chosen so that exploration would hold less importance while also having an impact on the results. Had epsilon been larger, the difference between Sarsa and Q-Learning would have been more evident. Decaying epsilon and learning rates would take unpredicted changes in the environment into account as the values would change the longer the algorithms run.

References

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Cliff Walking - Gymnasium Documentation. (n.d.). Retrieved from https://gymnasium.farama.org/environments/toy_text/cliff_walking/